Kalshiman

A small bot that watches the markets all night, so you don't have to.

The full breakdown - built on paper, version by version - 2026-08-20 04:22 UTC

A note before the numbers

Every fifteen minutes, Kalshi runs a market on gold, silver, or crypto: up or down in the next quarter hour. Most of the time, the market already knows the answer before the round is over. A side priced at 92 cents settles that way about 97% of the time. That isn't a secret. The secret is having the patience to wait for the 92-cent moment, the nerve to buy it, and the self-control to let it finish instead of flinching at the first dip.

The first version of this bot had none of that. It traded constantly, lost fifty cents in gaps it never saw coming, and never once asked what actually happened after a trade. It won 79% of the time and lost money anyway. We rebuilt it. The new one barely trades at all - and made more in one hour than the old one made in a day.

Here's the honest state of it: everything in this document is paper. The real account is untouched, and it stays that way until the paper record holds up over many days. What you're about to read is the story of a bot learning to keep its hands still - and the numbers that prove it.

The bigger idea is the one that keeps us going: what if your bot worked with other bots? A fleet that trades together, shares what it learns, and spreads the work so no single machine has to watch every market every night. Consistent, high-throughput, near-zero-cost trading. Every quiet day becomes free gains - and the humans get their evenings back. Kalshiman is the first bot in that fleet, and it just learned to stop losing money.

1. The 92-cent question

Walk up to a gold market fifteen minutes before it closes. The price is 92 cents for “up.” Ask a stranger what happens and they'll shrug. Ask the market itself, and it has already told you: the buyers have decided, and almost all of the time - 97% of the time on our measured record - that side is right.

So where's the catch? The catch is that 97% isn't 100%. The other 3% of the time, the 92-cent side collapses, and a loss costs you 92 cents - almost eleven times what a win pays. That's the whole game in one paragraph: win small, often. Lose small, rarely. Never let the two get out of balance.

The first version of Kalshiman got the first half right and the second half catastrophically wrong. It was a coin that landed heads often - and paid out like a lottery ticket when it landed tails.

We also learned that trading fails in exactly two directions. Too hot: you trust every hunch, chase every wiggle, and get stopped out on noise. Too cold: you wait for certainty that never comes, or let a formula stop you out of winners because it can't tell a dip from a disaster. The old bot was too cold in a specific way: its stop treated every dip like an emergency. Nine of its fifteen stopped trades recovered and won at the bell. The stop wasn't protecting anything. It was donating wins.

The fix turned out to be a balance, not a formula - three layers, each one catching the other two's failure mode. That's the subject of chapter four.

2. What we learned the hard way

Before the rebuild, the bot ran on real money for a few weeks. The record is not pretty, and we're not going to pretend it is. It's the reason everything after it exists.

What we measured	Value
Settled trades	291
Win rate	78.7%
Net P&L	-$77.85
Worst drawdown	-$91.43
Average win / average loss	+12.3c / -48.5c
Median loss	-100.7c (a full reversal)

The math is one line: with 12-cent wins and 48-cent losses, break-even needs a 134% win rate. The bot was winning at 79% and losing anyway. The edge was real; the losses were simply too big for the wins.

Three specific failures explain almost all of it:

- **Stops that didn't stop.** Paper trades gaped 17 to 65 cents past an 8-cent stop, because the stop only checked every five seconds and the market didn't wait for it.

- **Stops that ate winners.** A dip on a high-confidence pick is usually noise that recovers. Nine of fifteen stopped trades came back and won. We were selling winners to protect against a tail we were too small to fear.

- **No journal.** The bot never asked what happened after a trade. It couldn't learn from its own mistakes, so it repeated them.

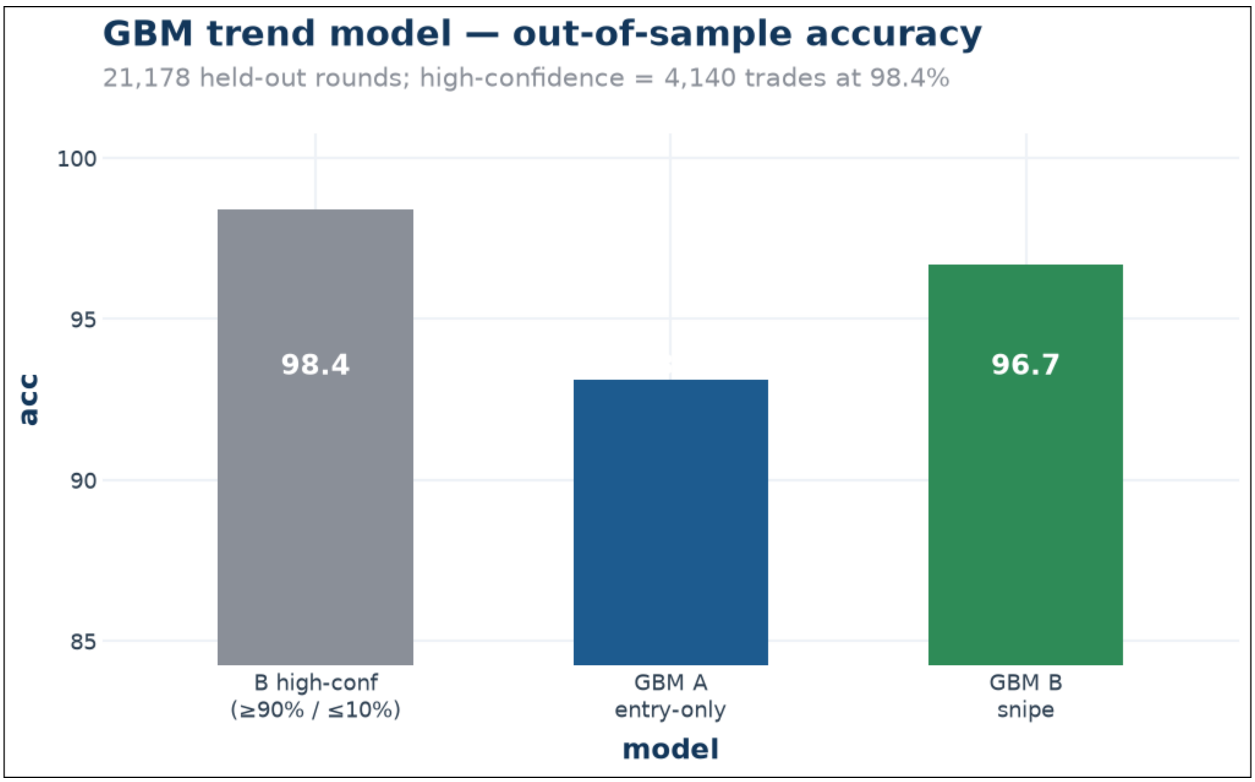
Gold was the leak in miniature: 110 trades, a 75% win rate, and an average loss of $5.46 - oversized positions and full reversals. The single-trade disasters were exact-spot markets, combos, and one hourly oil trade. All of them are banned now.

3. The rebuild: data before opinions

We stopped guessing and started measuring. We pulled every finished round we could find - 105,000 of them - and laid them out the way a trader sees them: what the market looked like before the entry, how it moved afterward, and where it landed. Then we audited the data for leaks, because a model that reads the answer before it answers isn't a model, it's a cheat sheet. The audit passed.

Out of that dataset we trained a gradient-boosted model. On rounds it had never seen:

Model	What it sees	Accuracy (out-of-sample)
Entry-only	the snapshot before you enter	93.1%
Snipe	plus the first two minutes of the trend	96.7%
Snipe, high-confidence	its strongest calls only	98.4% (4,140 trades)



The model's out-of-sample accuracy, strongest to strongest

We also built the honest evaluation loop: a dashboard that plays every round back, shows the model's reasoning, and scores it against reality. For the first time, we could watch the bot think - and see exactly where it was wrong.

4. How it works now: three layers

Watch a good trader for ten minutes and you'll notice they don't compute, they observe - the book, the candles, the way the market is moving - and they ask one question: does this look like the setups that win? Kalshiman is three people in one.

A statistician. It has read 105,000 finished rounds and knows, to the percentage point, how often a 92-cent gold side actually settles. When it says 98%, it's right 98.4% of the time. It is calibrated, and it is never surprised.

A quiet observer. A small language model reads each round the way a person would - the book, the path, the history - and gives a one-line opinion. It is often wrong on its own, which is exactly why it is never in charge. Its opinions are written down and scored later, like a trader's journal.

A disciplined hand. It only acts when the statistician is confident and the observer agrees. Once in, it does the hard part: it lets a dip breathe for three minutes before it even considers leaving, and if it does leave, it leaves at the stop level - never at whatever panicked price the market printed while it wasn't looking.

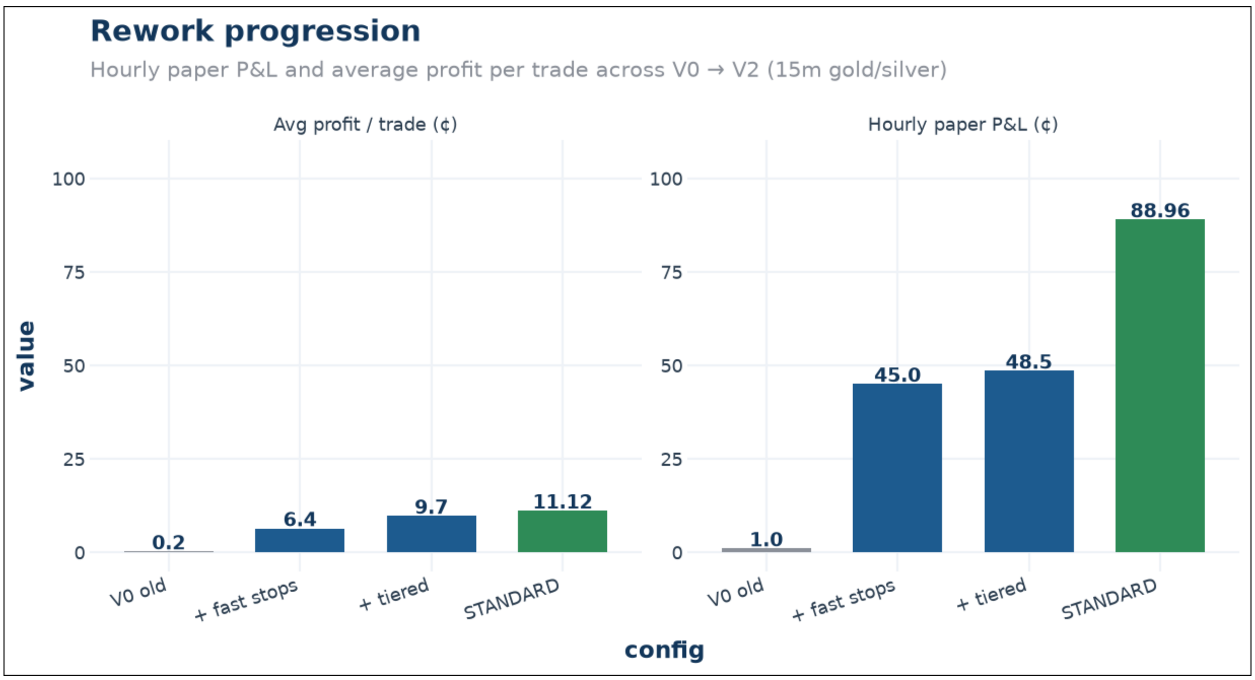
Step	What happens
Scan	gold15-late, silver15-late, universal snipe (0.5-2.5m to close), every 5 seconds
Score	the researcher pulls live candles and scores logistic + GBM + GBM-snipe
Gate	price band, timing, EV, liquidity - then the GBM bar: >=85% or <=15% and agreement
Enter	a paper trade, one contract
Manage	every tick, plus a fast check every 1.5 seconds, paper and live
Exit, win	bank at 99-99.5c, or hold to resolution
Exit, loss	tiered stop: 15c for entries >=90c, 8c below; a dip gets 3 minutes to recover; close at the stop level
After	a written post-mortem against the real result

Every rule in that table exists because a specific losing pattern taught it to us. The three-minute grace exists because nine of fifteen stops used to recover and win. The stop-level fill exists because a 15-cent stop once booked a 50-cent loss. This is a system that keeps its receipts.

5. The receipts

Here is the rework, hour by hour, on paper. Each row is the change that made the next row possible.

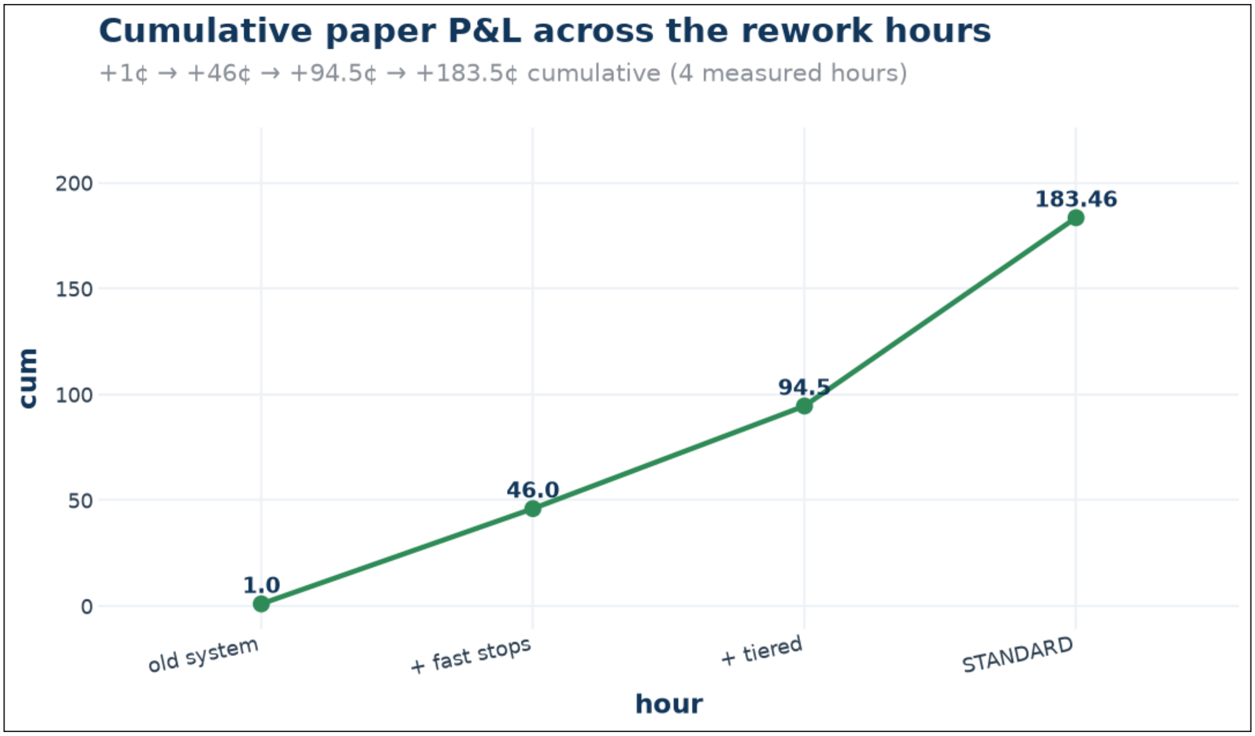
What changed	Trades	Result	Avg/trade
The old system	5	+1c, one loss	+0.2c
Paper stops that actually work	7	+45c, one loss	+6.4c
Stops that respect the entry	5	+48.5c, no losses	+9.7c
Stops that wait out noise	8	+88.96c, no losses	+11.12c



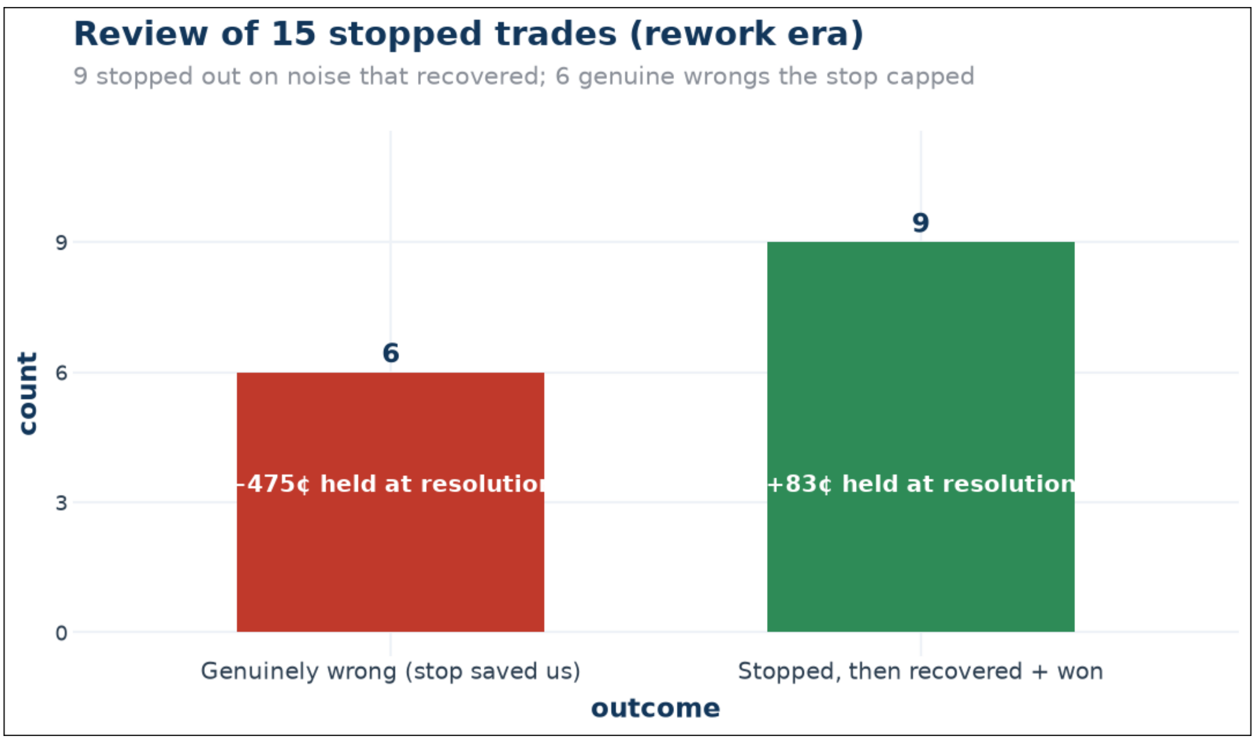
The same story as a chart: money per hour, and money per trade

The standard hour, trade by trade: gold NO at 90c banked 6.74c. Silver YES at 81c held to the bell for 23c. Gold NO at 93c banked 2.74c. Silver NO at 93c held for 8c. Gold YES at 87c banked 4.74c. Silver YES at 86c held for 16c. Gold NO at 93c banked 4.74c. Silver NO at 81c held for 23c.

No heroics. No last-second saves. The bot let the market do the work and got out of its way. That's the whole strategy, and it's boring on purpose.



Cumulative paper P&L; across those four hours



Why the stops changed: nine recovered and won, six were genuine wrongs

6. Risk, and the stop policy

A stop is a promise you make to yourself before the trade: if this goes wrong, it costs me this much and not a cent more. The old bot broke that promise in both directions - it fired on noise, and when a real break happened it filled at whatever the market panicked to. The new policy fixes both.

Control	What it does
Tiered stops	entry >=90c gets 15c of room; below that, 8c - high prices mean the dip is almost always noise
Recovery grace	a dip has 3 minutes to prove it's real; if the market comes back, the winner stays
Limit-stop fills	the exit is marked at the stop level - a crash price is never booked
Fast checks	every 1.5 seconds, paper and live
Hard bans	no martingale, no revenge trading, no leverage, no both-sides arbs
Real money	off until the paper record earns it

The math that justifies it: of the old system's fifteen stops, nine recovered and won (the flat stop cost us 188c of forgone wins) and six were genuine wrongs (the stop saved 344c). Net, stopping still saved money - the problem was never stopping, it was stopping on noise and filling at crash prices.

7. What it could mean - the honest compounding story

The unit of edge is a percent, not a cent. Buy the favorite at 90 cents, it settles at 100: that is +11.1% on what you deployed. The 11 cents in our reports is the per-contract unit; the dollars you actually make scale with position size.

Why growth accelerates: the return is a percentage of your stake, and your stake is a percentage of your balance. At a $200 balance and a $50 entry, ten perfect trades are +$55.60 - the resolution stops making cents and starts making dollars. At a 25% stake ratio and a 93% win rate, balance grows about +2.4% per trade, so $200 to $500 takes roughly 39 trades while the first dollars took hundreds. The early phase is fragile (minimum contracts force near-all-in below $4), which is exactly why real size waits until $200-300 is banked as backing - and why the stake ratio stays at ~25% so stops and the rare full loss stay survivable. This is the sizing ladder, and the paper test is still at rung one: $1.

Balance	Stake (25%)	One win (+11%)	One stop
$5	$1.25	+$0.14	-$0.11
$30	$7.50	+$0.83	-$0.67
$100	$25	+$2.78	-$2.22
$200	$50	+$5.56	-$4.44
$500	$125	+$13.89	-$11.11

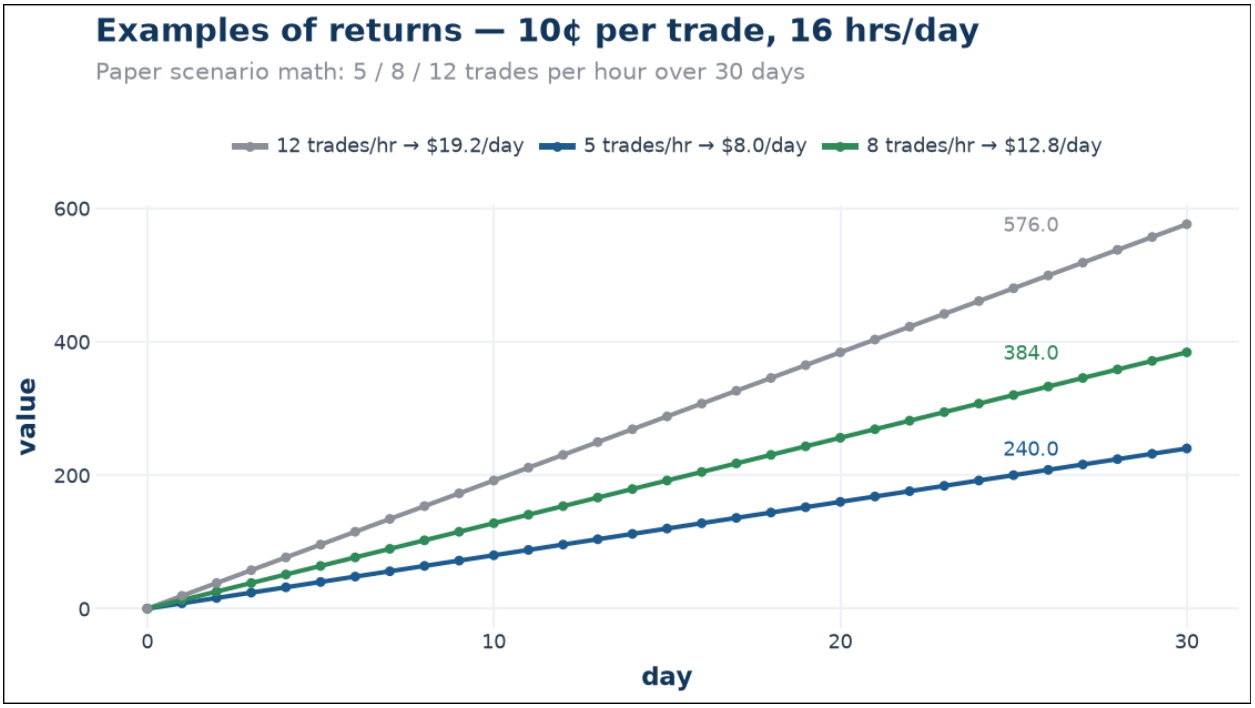
Position	One win (+11%)	One stop (-8c on a 90c entry)
$1	+$0.11	-$0.09
$50	+$5.56	-$4.44
$150	+$16.67	-$13.33
$500	+$55.56	-$44.44

The compounding math, done honestly. Reinvest everything and the per-trade multiplier is W x 1.111 + L x (loss). At 98% wins with 8c stops that is +10.7% per trade - $1 reaches $200 in about 53 trades and $500 in 62. At 64 trades a day, that is under a day of perfect compounding. Which is the truth of the whole thing: the math is not the bottleneck. Win rate and liquidity are.

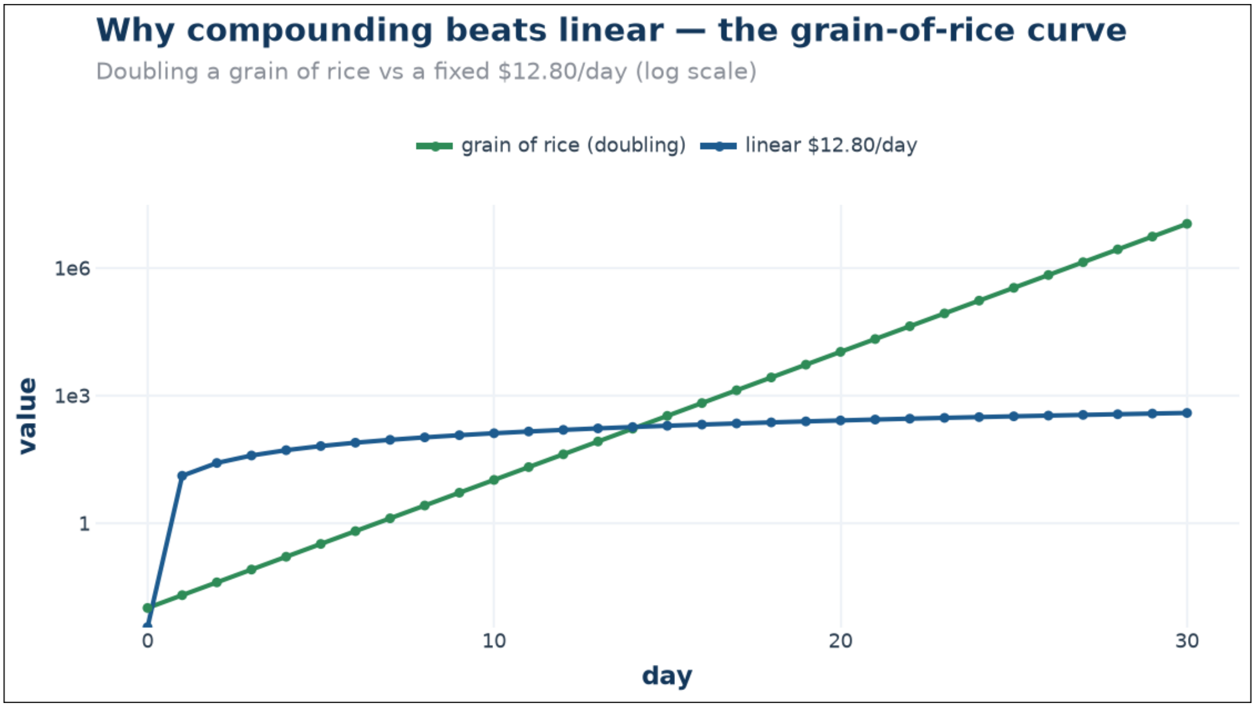
Win rate	Loss shape	Growth/trade	Trades to $200	Trades to $500

98%	8c stop	+10.7%	53	62
96%	8c stop	+9.6%	59	69
96%	full loss (hold wrong)	+6.7%	83	97

Why it does not go to infinity: win rate is not 100% (a full-loss final-minute hold drags the multiplier to +6.7%), and liquidity is a wall - a 15m round has maybe $50-100k of depth, so a $500 position is fine and a $50,000 one destroys the edge. The fleet is how this scales: not one huge position, but many workers, many markets, sharing what they learn. That is the pool.



Flat per-trade scenario math - before position sizing



Why compounding beats linear - the grain-of-rice curve

8. The pool - how a fleet works

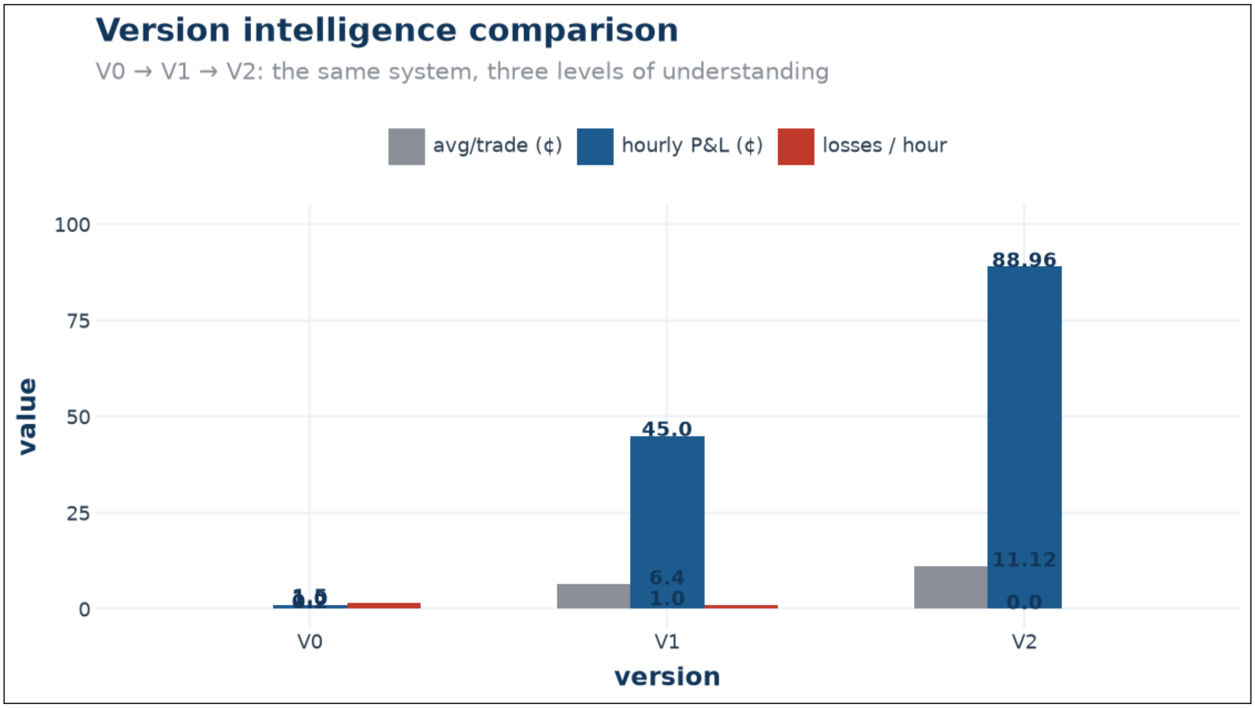
One bot can watch a handful of markets every night. Ten bots can watch all of them, and each one gets to be lazier because the others cover the rest. The pool is the nervous system that lets workers share what they find instead of each learning everything alone.

Workers publish high-confidence snapshots to a shared inbox; other workers verify them and earn credits; the poster is refunded when the information proves actionable. Publishing is capped at one post per 10 minutes, verification is open at any time, and the main node is the only one that pushes verified strategies down. Workers naturally specialize - a node in another timezone covers the overnight markets, a commodity-only node runs just those lanes - and because everyone shares for a flat, small rate, the fleet converges toward the same knowledge over time.

Current status, honestly: the pool is disabled during the paper test. The live worker burned its daily KV quota through sync fan-out, not trading (fixed client-side; a worker cache is staged and needs deploying from the account that owns the pool). The roadmap is deploy the cache, re-enable, run two nodes on paper side by side, then scale positions within each market's liquidity - not beyond it.

9. Where we're going

The bot didn't get smarter by adding parameters. It got smarter by learning from its own outcomes, refusing weak setups, and finally understanding the difference between noise and signal. That's not magic - it's a trader who started keeping a journal.



V0 to V2: the same system, three levels of understanding

What's next is boring in the best way: more markets with trained models, a slightly earlier entry window on the setups the model is sure of, and day after day of net-positive paper hours. Then, only then, real money - and the first worker in a fleet that trades together.

One perfect hour isn't a track record. We know that. But for the first time since we started, the bot isn't the reason we're not there yet.

10. Straight answers

What is the actual edge?

Ultra-short markets converge: a side implied at 90%+ settles that way ~96-99% of the time. We enter late, with a confident model and an observer's agreement, ride winners through noise, and cap the rare wrong call at a fixed stop level.

Why did the old system fail?

It was a static formula with no sense of context, stops that gaped 17-65c on paper, and no journal. The rework fixed all three.

What is the LLM for?

It's the observer - it reads each round like a trader would and gives a one-line opinion. Advisory only, logged, scored, never in charge.

What are realistic returns?

Paper standard hour: 8W/0L, +88.96c, 11.12c average, $0 drawdown. At that pace, roughly $12-14 a day paper.

What are the biggest risks?

A real reversal on a high-price entry (capped at the stop), a regime change that breaks the pattern until the journal catches it, Kalshi fee or availability changes - and one perfect hour not being a track record.

Does it use leverage?

No. No martingale, no revenge trading, no leverage, no both-sides. Real money is off until the paper record earns it.

Why run it at all?

A few watts for a self-correcting system with a quantified edge and capped downside - and the first worker in a fleet that makes quiet days into free gains.

Companion markdown: docs/00-thesis.md, docs/13-versions.md, docs/14-reasoning.md, docs/15-returns.md, ARCHITECTURE.md. Charts: docs/charts/*.png (plotnine).